Proposition 1.6. *The second-order linear PDEs can always be written as*

$$a(x,y)u_{xx} + b(x,y)u_{xy} + c(x,y)u_{yy} + d(x,y)u_x + e(x,y)u_y + f(x,y)u = g(x,y)$$

$$\begin{cases} 4ac - b^2 > 0 & \textit{elliptic} \\ 4ac - b^2 = 0 & \textit{parabolic} \\ 4ac - b^2 < 0 & \textit{hyperbolic} \end{cases}$$

Theorem 1.10. *Separable ODE can be solved in the following way*

$$y' + p(x)y = 0 \quad\Rightarrow\quad \frac{dy}{dx} + p(x)y = 0 \quad\Rightarrow\quad \frac{dy}{y} = -p(x)dx \quad\Rightarrow\quad \int \frac{dy}{y} = -\int p(x)dx$$

and the solution is

$$u(x) = Ce^{-\int p(x)dx} \tag{1.11}$$

Theorem 1.12. *Linear ODE can be solved by the following procedure*

1. *Solve the corresponding separable equation $y' - p(x)y = 0$ to obtain a solution $\hat{y} = e^{\int p(x)dx}$.*
2. *Multiply the linear ODE by $\hat{y}$ and rewrite the ODE*

$$y' + p(x)y = q(x) \quad\Rightarrow\quad \hat{y}(y' + p(x)y) = \hat{y}q(x) \quad\Rightarrow\quad (\hat{y}y)' = \hat{y}q(x)$$

3. *Integrate the above equation*

$$\hat{y}y = \int \hat{y}q(x)dx + C \quad\Rightarrow\quad y = \frac{1}{\hat{y}}\left(\int \hat{y}q(x)dx + C\right)$$

$$\Rightarrow\quad y = e^{-\int p(x)dx}\left(\int q(x)e^{\int p(x)dx}dx + C\right)$$

Definition 1.14 (Second order ODEs)**.** *The constant coefficient second order ODEs are the following equations*

$$ay'' + by' + cy = 0 \ (a \neq 0). \tag{1.13}$$

Theorem 1.15. *Constant coefficient second order ODEs can be solved by the following procedure*

1. *Solve the characteristic equation $a\lambda^2 + b\lambda + c = 0$ to get two solutions λ_1 and λ_2.*
2. *If $\lambda_1 \neq \lambda_2$, the general solution is*

$$y(x) = C_1e^{\lambda_1 x} + C_2e^{\lambda_2 x} \tag{1.14}$$

3. *If $\lambda_1 = \lambda_2 = \lambda$, the general solution is*

$$y(x) = (C_1 + C_2x)e^{\lambda x} \tag{1.15}$$

4. *If λ_1, λ_2 are complex roots $\alpha \pm i\beta$, apply the Euler's formula to rewrite (1.14)*

$$y(x) = C_1e^{(\alpha+i\beta)x} + C_2e^{(\alpha-i\beta)x} = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) \tag{1.16}$$

where $c_1 = C_1 + C_2$, $c_2 = i(C_1 - C_2)$.

1. $$\int_0^L x\sin\left(\frac{n\pi x}{L}\right)dx = \frac{L^2}{n\pi}(-1)^{n+1}$$
2. $$\int_0^L x^2\sin\left(\frac{n\pi x}{L}\right)dx = \left(-\frac{L^3}{n\pi} + \frac{2L^3}{(n\pi)^3}\right)(-1)^n - \frac{2L^3}{(n\pi)^3}$$
3. $$\int_0^L x\cos\left(\frac{n\pi x}{L}\right)dx = \begin{cases} 0, & \text{当 } n \text{ 为偶数} \\ -2\left(\frac{L}{n\pi}\right)^2, & \text{当 } n \text{ 为奇数} \end{cases}$$
4. $$\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)dx = \frac{2L^3}{(n\pi)^2}(-1)^n$$
5. $$\int_0^L e^x\sin\left(\frac{n\pi x}{L}\right)dx = \frac{n\pi L}{L^2 + (n\pi)^2}\left((-1)^{n+1}e^L + 1\right)$$

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\sin A\sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$$

$$\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$$

嗟:

$$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

Laplace's Eq

$$X''Y + XY'' = 0 \quad\Rightarrow\quad \frac{X''}{X} = -\frac{Y''}{Y}$$

$$X'' + \lambda X = 0, \quad Y'' - \lambda Y = 0.$$

For $\lambda > 0$, by writing $\lambda = k^2 (k > 0)$ we have

$$u(x,y) = \left(A_1e^{ikx} + A_2e^{-ikx}\right)\left(B_1e^{ky} + B_2e^{-ky}\right),$$

and for $\lambda < 0$, by writing $\lambda = -l^2 (l > 0)$ we have

$$u(x,y) = \left(A_1e^{lx} + A_2e^{-lx}\right)\left(B_1e^{ily} + B_2e^{-ily}\right).$$

Therefore, we get

$$u = \begin{cases} (A_1x + A_2)(B_1y + B_2), \\ \left(A_1e^{ikx} + A_2e^{-ikx}\right)\left(B_1e^{ky} + B_2e^{-ky}\right), \\ \left(A_1e^{lx} + A_2e^{-lx}\right)\left(B_1e^{ily} + B_2e^{-ily}\right). \end{cases}$$

Instead of (1.23), we can also choose

$$X = \cos(\sqrt{\lambda}x), \sin(\sqrt{\lambda}x), \quad Y = \cosh(\sqrt{\lambda}y), \sinh(\sqrt{\lambda}y).$$

In this case, we have

$$u(x,y) = (A_1\cos(kx) + A_2\sin(kx))(B_1\cosh(ky) + B_2\sinh(ky))$$

$$u(x,y) = (A_1\cosh(lx) + A_2\sinh(lx))(B_1\cos(ly) + B_2\sin(ly))$$

$$\int_{-L}^{L}\cos\frac{n\pi x}{L}\cos\frac{m\pi x}{L}dx = \begin{cases} 2L & (n = m = 0), \\ L\delta_{nm} & (\textit{otherwise}), \end{cases}$$

$$\int_{-L}^{L}\sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}dx = \begin{cases} 0 & (n = m = 0), \\ L\delta_{nm} & (\textit{otherwise}), \end{cases}$$

$$\int_{-L}^{L}\sin\frac{n\pi x}{L}\cos\frac{m\pi x}{L}dx = 0 \quad (\textit{all } n, m).$$

$$\int^{L} e^{in\pi x/L}e^{-im\pi x/L}dx = \int_{-L}^{L}e^{i(n-m)\pi x/L}dx = 2L\delta_{mn}.$$

$$\frac{\pi x}{L}\cos\frac{m\pi x}{L}dx = \begin{cases} L & (n = m = 0), \\ \frac{L}{2}\delta_{nm} & (\textit{otherwise}), \end{cases}$$

$$\frac{\pi x}{L}\sin\frac{m\pi x}{L}dx = \begin{cases} 0 & (n = m = 0), \\ \frac{L}{2}\delta_{nm} & (\textit{otherwise}), \end{cases}$$

Fouier Series, f: [-L, L] to R

$$f(x) = \sum_{n=1}^{\infty}B_n\sin\frac{n\pi x}{L}, \quad 0 < x < L,$$

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}dx$$

$$f(x) = A_0 + \sum_{n=1}^{\infty}\left(A_n\cos\frac{n\pi x}{L} + B_n\sin\frac{n\pi x}{L}\right)$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L}f(x)dx,$$

$$A_n = \frac{1}{L}\int_{-L}^{L}f(x)\cos\frac{n\pi x}{L}dx,$$

$$B_n = \frac{1}{L}\int_{-L}^{L}f(x)\sin\frac{n\pi x}{L}dx.$$

$$f(x) = A_0 + \sum A_n\cos\frac{n\pi x}{L}, \quad 0 < x < L,$$

$$A_0 = \frac{1}{L}\int_0^L f(x)dx, \quad A_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}dx$$

Euler's formula $e^{ix} = \cos x + i\sin x$

$$f(x) = \sum_{n=-\infty}^{\infty}\alpha_ne^{in\pi x/L},$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \qquad \sin x = \frac{e^{ix} - e^{-ix}}{2i}. \qquad \alpha_n = \frac{1}{2L}\int_{-L}^{L}f(x)e^{-in\pi x/L}dx$$

Parseval's Thm for pw smooth function

$$\frac{1}{2L}\int_{-L}^{L}f(x)^2dx = A_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}\left(A_n^2 + B_n^2\right) \qquad \sigma_N^2 = \frac{1}{2L}\int_{-L}^{L}[f(x) - f_N(x)]^2dx$$

$$\sigma_N^2 = \frac{1}{2}\sum_{n=N+1}^{\infty}\left(A_n^2 + B_n^2\right)$$

Theorem 2.22 (Integral test)**.** *Given a monotonic and positive function $f(x)$, we have*

$$\int_{N+1}^{\infty}f(x)dx \leq \sum_{n=N+1}^{\infty}f(n) \leq \int_N^{\infty}f(x)dx$$

Parseval's Thm: complex

$$\frac{1}{2L}\int_{-L}^{L}|f(x)|^2dx = \sum_{n=-\infty}^{\infty}|\alpha_n|^2.$$

Parseval's Thm: Fourier sine

$$\frac{1}{L}\int_0^L f(x)^2dx = \frac{1}{2}\sum_{n=1}^{\infty}B_n^2$$

Solving 1st order PDE by MoC

linear (no u)

linear homo

ex: $a = x+1$, $b = -(y+1)$

$$\begin{cases} a(x,y)u_x + b(x,y)u_y = 0 \\ u(x,0) = f(x) \end{cases}$$

homo

ex: $f(x) = x+1$

Sol **step1** Characteristic curve

$$\begin{cases} \frac{d\tilde{x}}{ds} = a(\tilde{x},\tilde{y}) \quad (\tilde{x}+1) \\ \frac{d\tilde{y}}{ds} = b(\tilde{x},\tilde{y}) \quad (-(\tilde{y}+1)) \end{cases} \Rightarrow \begin{cases} \tilde{x}(s) = \ldots \ (-1+Ae^s), & \tilde{x}(0) = x_0 \\ \tilde{y}(s) = \ldots \ (-1+Be^{-s}), & \tilde{y}(0) = 0 \end{cases}$$

$\int\frac{1}{y+1} = \int 1$

$-\ln|y+1| = -s$

$y+1 = Be^{-s}$

$A = x_0 + 1$

$B = 1$

step2 for (x,y) on the curve

$$\begin{cases} x = x(s_1) \quad -1+(x_0+1)e^{s_1} \\ y = y(s_1) \quad -1+e^{-s_1} \end{cases} \Rightarrow x_0 = \ldots(x,y)$$

$(x+1)(y+1) = x_0 + 1$

$\Rightarrow x_0 = (x+1)(y+1) - 1$

$$u(x,y) = u(x_0,0) = f(x_0) = f(\ldots(x,y))$$

$= x_0 + 1 = (x+1)(y+1)$

quasi-linear

ex: u^2, 1, u

$$\begin{cases} a(x,y,u)u_x + b(x,y,u)u_y = c(x,y,u) \\ u(x,V(x)) = f(x) \end{cases}$$

ex $u(x,0) = x$

step1

$$\begin{cases} \frac{d\tilde{x}}{ds} = a \quad (\tilde{u}^2) \\ \frac{d\tilde{y}}{ds} = b \quad (1) \\ \frac{d\tilde{u}}{ds} = c \quad (\tilde{u}) \end{cases} \begin{cases} \tilde{x}(0) = x_0 \\ \tilde{y}(0) = y_0 \quad (0) \\ \tilde{u}(0) = f(x_0) \quad (x_0) \end{cases} \Rightarrow \begin{cases} \tilde{x}(s) = x_0 + \frac{1}{2}x_0^2(e^{2s} - 1) \\ \tilde{y}(s) = s \\ \hat{u}(s) = x_0e^s \end{cases}$$

on curve

$$\begin{cases} x = \tilde{x}(s_1) \\ y = \tilde{y}(s_1) \\ u = \hat{u}(s_1) \end{cases} \Rightarrow \begin{cases} x = x_0 + \frac{1}{2}x_0^2(e^{2s_1} - 1) \\ y = s_1, \quad u = x_0e^{s_1} \end{cases}$$

$$\Rightarrow x_0^2 + \frac{2}{e^{2y}-1}x_0 - \frac{1}{e^{2y}-1}x = 0$$

$$x_0 = \frac{-2(e^{2y}-1) + \sqrt{\frac{4}{(e^{2y}-1)^2} + 4\frac{x}{e^{2y}-1}}}{2} \quad \left(\frac{-b \pm \sqrt{\Delta}}{2a}, \text{ judge } \pm\right)$$

$$x_0 = \frac{-1 + \sqrt{1 + x(e^{2y}-1)}}{e^{2y}-1}$$

$u = x_0e^{s_1}$

$= \sim e^y$

Thm Let $\tilde{u}(s) = u(\tilde{x},\tilde{y}) \Rightarrow \frac{d\tilde{u}}{ds} = c(\tilde{x},\tilde{y},\tilde{u})$

Pf $\frac{d\tilde{u}}{ds} = u_x\frac{d\tilde{x}}{ds} + u_y\frac{d\tilde{y}}{ds} = u_xa + u_yb = c(\tilde{x},\tilde{y},u(\tilde{x},\tilde{y}))$

Question 4. Solve $u_t + uu_x = u$, $-\infty < x < \infty$, $t > 0$ with the initial condition $u(x,0) = x^2$, $-\infty < x < \infty$. using the method of characteristics.

Sol $u_t + uu_x = u$

Characteristic curves

$$\begin{cases} \frac{d\tilde{x}}{ds} = \tilde{u}, & \tilde{x}(0) = x_0 \\ \frac{d\tilde{t}}{ds} = 1, & \tilde{t}(0) = 0 \\ \frac{d\tilde{u}}{ds} = \tilde{u}, & \tilde{u}(0) = x_0^2 \end{cases} \Rightarrow \begin{cases} \tilde{t}(s) = s + B, \ B = 0 \\ \tilde{u}(s) = Ce^s, \ C = x_0^2 \end{cases}$$

$$\Rightarrow \begin{cases} \tilde{t}(s) = s \\ \tilde{u}(s) = x_0^2e^s \end{cases}$$

So $\frac{d\tilde{x}}{ds} = x_0^2e^s \Rightarrow \tilde{x}(s) = \int_0^s x_0^2e^ydy + A$

$= x_0^2e^s + A$

$\tilde{x}(0) = x_0 \Rightarrow A = x_0 - x_0^2$

So $\tilde{x}(s) = x_0^2(e^s - 1) + x_0$

On curve: $x = \tilde{x}(s_1)$, $t = \tilde{t}(s_1)$

$= x_0^2(e^{s_1} - 1) + x_0 \quad = s_1$

$= x_0^2(e^t - 1) + x_0 \Rightarrow (e^t - 1)x_0^2 + x_0 - x = 0$

$$x_0 = \frac{-1 - \sqrt{1 + 4(e^t - 1)}}{2(e^t - 1)}$$

since $x_0 \to \infty$ if $t \to \infty$ if t

So $u(x,t) = \hat{u}(s_1) = x_0^2e^s = \left(\frac{-1 - \sqrt{1 + 4(e^t-1)}}{2(e^t - 1)}\right)^2e^t$

Solve homogeneous heat eq:

$$\begin{cases} u_t = K u_{zz}, & 0 < z < L, \quad t > 0, \\ u \cos\alpha - L u_z \sin\alpha = 0, & z = 0, \quad t > 0, \\ u \cos\beta + L u_z \sin\beta = 0, & z = L, \quad t > 0, \\ u = f(z), \quad T_\lambda = C_n e^{-K\lambda_n t} & 0 < z < L, \quad t = 0, \end{cases}$$

$$T'(t) + \lambda K T(t) = 0, \qquad \underbrace{Z''(z) + \lambda Z(z) = 0, \quad Z(0)\cos\alpha - LZ'(0)\sin\alpha = 0, \quad Z(L)\cos\beta + LZ'(L)\sin\beta = 0.}_{SL}$$

Solve for Z_n, λ_n

$$\Rightarrow u(z,t) = \sum_{n=1}^{\infty} C_n Z_n e^{-K\lambda_n t}$$

And to match $u(z,0) = f(z)$

$$\Rightarrow f(z) = \sum_{n=1}^{\infty} C_n Z_n$$

orthogonal, can expand

$$C_n = \frac{\int_0^L f(z) Z_n \, dz}{\int_0^L Z_n^2 \, dz}$$

Solve time-ind nonhomogeneous heat eq

$$\begin{cases} u_t = K u_{zz} + r(z) \\ u\cos\alpha - L u_z \sin\alpha = T_1, \ z=0 \\ u\cos\beta + L u_z \sin\beta = T_2, \ z=L \\ u = f(z) \end{cases}$$

Step 1 Solve $U(z)$ for

$$\begin{cases} KU'' + r(z) = 0 \\ U\cos\alpha - LU'\sin\alpha = T_1, \ z=0 \\ U\cos\beta + LU'\sin\beta = T_2, \ z=L \end{cases}$$

2. Define $V(z,t) = u(z,t) - U(z)$

Solve homogeneous

$$\begin{cases} V_t = K_{zz} \\ V\cos\alpha - LV_z \sin\alpha = 0, \ z=0 \\ V\cos\beta - LV_z \sin\beta = 0, \ z=L \\ V(z,0) = f(z) - U(z), \ t=0 \end{cases}$$

Step 3 $u(z,t) = V(z,t) + U(z)$

The formula way to solve inhomogeneous heat eq

$$\begin{cases} u_t = K u_{zz} + r(z,t), & t > 0, \quad 0 < z < L, \\ u(0,t)\cos\alpha - L u_z(0,t)\sin\alpha = T_1(t), & t > 0, \\ u(L,t)\cos\beta + L u_z(L,t)\sin\beta = T_2(t), & t > 0, \\ u(z,0) = f(z), & 0 < z < L, \end{cases}$$

Step 1:

We first solve the problem below. $U(z,t) = A(t) + B(t)z$

$$\begin{cases} U_{zz}(z,t) = 0, & 0 < z < L, \\ U(0,t)\cos\alpha - LU_z(0,t)\sin\alpha = T_1(t), & t > 0, \\ U(L,t)\cos\beta + LU_z(L,t)\sin\beta = T_2(t), & t > 0. \end{cases}$$

Step 2:

We define $v(z,t) = u(z,t) - U(z,t)$. We also introduce

$$R(z,t) = r(z,t) - U_t(z,t), \quad F(z) = f(z) - U(z,0).$$

Thus, the new problem for $v(z,t)$ becomes

$$\begin{cases} v_t = K v_{zz} + R(z,t), & t > 0, \quad 0 < z < L, \\ v(0,t)\cos\alpha - L v_z(0,t)\sin\alpha = 0, & t > 0, \\ v(L,t)\cos\beta + L v_z(L,t)\sin\beta = 0, & t > 0, \\ v(z,0) = F(z), & 0 < z < L. \end{cases}$$

Step 3:

We solve (2.19) for $v(z,t)$, and finally we obtain $u(z,t) = U(z,t) + v(z,t)$.
To solve it (2.19), We consider the following Sturm-Liouville problem:

$$\begin{cases} \phi'' + \lambda\phi = 0, & 0 < z < L, \\ \phi(0)\cos\alpha - L\phi'(0)\sin\alpha = 0, \\ \phi(L)\cos\beta + L\phi'(L)\sin\beta = 0. \end{cases}$$

Let λ_n and ϕ_n be eigenvalues and eigenfunctions of this Sturm-Liouville eigenproblem. We express v, R, F with ϕ_n (recall the convergence theorem):

$$v(z,t) = \sum_{n=1}^{\infty} v_n(t)\phi_n(z), \quad R(z,t) = \sum_{n=1}^{\infty} R_n(t)\phi_n(z), \quad F(z) = \sum_{n=1}^{\infty} F_n\phi_n(z).$$

$$R_n = \frac{\int_0^L [r(z,t) - U_t(z,t)]\phi_n(z)dz}{\int_0^L \phi_n(z)^2 dz}, \quad F_n = \frac{\int_0^L [f(z) - U(z)]\phi_n(z)dz}{\int_0^L \phi_n(z)^2 dz}.$$

By substituting these expansions in (2.19), we obtain

$$\begin{cases} v_n'(t) = -\lambda_n K v_n(t) + R_n(t), & t > 0, \\ v_n(0) = F_n. \end{cases}$$

We can solve this equation as (recall we can always solve first-order ODEs):

$$v_n(t) = F_n e^{-\lambda_n K t} + \int_0^t R_n(s) e^{-\lambda_n K(t-s)} ds.$$

$$f(x,y) = \sum_{n=1}^{\infty} B_{mn} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2},$$

$$B_{mn} = \frac{4}{L_1 L_2} \int_0^{L_2} \int_0^{L_1} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2} dx dy.$$

SL problem:

$$[s(x)\phi'(x)]' + [\lambda\rho(x) - q(x)]\phi(x) = 0, \quad a < x < b, \qquad (3.26)$$

$$\phi(a)\cos\alpha - L\phi'(a)\sin\alpha = 0,$$

$$\phi(b)\cos\beta + L\phi'(b)\sin\beta = 0$$

Define the operator A by

$$A\phi = -\frac{1}{\rho(x)}\left([s(x)\phi'(x)]' - q(x)\phi(x)\right)$$

with domain

$$\left\{ \phi(x),\ x \in [a,b] \ \middle|\ \begin{array}{l} \phi(a)\cos\alpha - L\phi'(a)\sin\alpha = 0, \\ \phi(b)\cos\beta + L\phi'(b)\sin\beta = 0 \end{array} \right\}$$

then (3.26) is equivalent to $A\phi = \lambda\phi$.

Transfer any 2nd order linear ODE into SL:

$$a(x)y'' + b(x)y' + c(x)y$$

1. *Divide both sides by $a(x)$ to obtain $y'' + p(x)y' + q(x)y = 0$.*
2. *Solve the equation $\widehat{y}' = p(x)\widehat{y}$.*
3. *Multiply both sides of $y'' + p(x)y' + q(x)y = 0$ by $\widehat{y}$*

$$\widehat{y}(y'' + p(x)y') + \widehat{y}q(x)y = 0 \quad \Rightarrow \quad (\widehat{y}y')' + \widehat{y}q(x)y = 0$$

SL Thm: In SL problem,

- A is symmetric op; eigenvalues are real;
- eigenfunctions of different eigens are orthogonal;
- eigenfunctions of one eigen 互为倍数 (eigen multiplicity =1)

Generalized Fourier series:

Theorem 3.17 (Generalized Fourier series). *Given a Sturm-Liouville operator A of the form* (3.27) *with coefficient $\rho(x)$, let $\{\phi_n\}_{n\in\mathbb{N}} \subseteq Dom(A)$ be its eigenfunctions and $f(x) \in Dom(A)$ be an arbitrary piecewise smooth function. Then we have the following expansion*

$$f(x) = \sum_{n=1}^{\infty} c_n\phi_n(x), \qquad c_n = \frac{\langle f, \phi_n\rangle_\rho}{\|\phi_n\|^2} \qquad (3.55)$$

$$\langle \varphi, \psi\rangle_\rho = \int_a^b \varphi(x)\psi(x)\rho(x)dx$$

Generalized Parsevels:

$$\sigma_N^2 = \sum_{n=N+1}^{\infty} |c_n|^2 \|\phi_n\|^2 \qquad \|f\|^2 = \sum_{n=1}^{\infty} |c_n|^2 \|\phi_n\|^2$$

Solve homogeneous Wave eq

$$\begin{cases} U_{tt} = c^2(U_{xx} + U_{yy}), \quad 0<x<L_1, \ 0<y<L_2, \ t>0 \\ u = 0, \quad x = 0/L_1, \ y = 0/L_2 \\ u(x,y,0) = 0, \ U_t(x,y,0) = xy \end{cases}$$

$$f(x,y) = \sum_{n=1}^{\infty} B_{mn} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2},$$

$$\int_0^{L_2}\int_0^{L_1} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2} \sin\frac{m'\pi x}{L_1} \sin\frac{n'\pi y}{L_2} dxdy = \frac{L_1 L_2}{4}\delta_{mm'}\delta_{nn'}$$

$$B_{mn} = \frac{4}{L_1 L_2} \int_0^{L_2} \int_0^{L_1} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2} dx dy.$$

Step 1 $u = XYT$

$$\frac{T''}{T} = c^2 \frac{X''}{X} + \frac{Y''}{Y}$$

$$\frac{T''}{T} = -\lambda c^2, \quad \frac{X''}{X} = -\mu_1, \quad \frac{Y''}{Y} = -\mu_2$$

$$\lambda = \mu_1 + \mu_2$$

Step 2 $X(0) = X(L_1) = 0 \Rightarrow X_m = \sin\frac{m\pi x}{L_1}, \ \mu_m = \left(\frac{m\pi}{L_1}\right)^2$

Same for Y

$$\Rightarrow T = A\cos(\sqrt{\lambda}\, ct) + B\sin(\sqrt{\lambda}\, ct)$$

$$u = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2} \left(A_{mn}\cos(w_{mn}t) + B_{mn}\sin(w_{mn}t)\right)$$

$$w_{mn} = c\sqrt{\left(\frac{m\pi}{L_1}\right)^2 + \left(\frac{n\pi}{L_2}\right)^2}$$

$$u(x,y,0) = 0 \Rightarrow A_{mn} = 0$$

$$u_t(x,y,0) = xy \Rightarrow xy = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} B_{mn} w_{mn} \sin\frac{m\pi x}{L_1} \sin\frac{n\pi y}{L_2}$$

$$B_{mn} w_{mn} = \frac{4}{L_1 L_2} \int_0^{L_1} x \sin\frac{m\pi x}{L_1} dx \int_0^{L_2} y \sin\frac{n\pi y}{L_2} dy$$

5.1 Fourier trans

$$f(x)=\sum_{n=-\infty}^{\infty}\alpha_n e^{i\frac{n\pi}{L}x},\quad \alpha_n=\frac{1}{2L}\int_{-L}^{L}f(x)e^{-i\frac{n\pi}{L}x}\quad \to \mu_n,\ \Delta\mu_n=\frac{\pi}{L}$$

Let $\tilde f_L(\mu_n)=\frac{\alpha_n}{\pi/L}=\frac{1}{2\pi}\int_{-L}^{L}f(x)e^{-i\mu x}dx$

$$\Rightarrow f(x)=\sum_{n=-\infty}^{\infty}\tilde f_L(\mu_n)e^{i\mu_n x}\Delta\mu_n\qquad \boxed{L\to\infty\Rightarrow f(x)=\int_{-\infty}^{\infty}\tilde f(\mu)e^{i\mu x}d\mu}$$

where $\boxed{\hat f(\mu)=\frac{1}{2\pi}\int_{-\infty}^{\infty}f(x)e^{-i\mu x}dx}$

Parseval: $\frac{1}{2\pi}\int_{-\infty}^{\infty}|f(x)|^2dx=\int_{-\infty}^{\infty}|\hat f(\mu)|^2d\mu$

ex
$$f(x)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-m)^2}{2\sigma^2}},\quad \hat f(\mu)=\frac{1}{2\pi}e^{-im\mu}e^{-\frac{\mu^2\sigma^2}{2}}$$

$$f(x)=1=\lim_{\sigma\to\infty}e^{-\frac{x^2}{2\sigma^2}},\quad \hat f(\mu)=\delta(\mu)=\begin{cases}\infty, & \mu=0\\ 0, & \mu\neq 0.\end{cases}$$

$\int_{-\infty}^{\infty}\delta(x)\varphi(x)dx=\varphi(0)$ 在积分中选出 $x=0$处值

5.3 method of image

$$\begin{cases}u_t=ku_{xx}\\ u(0,t)=0\\ u(x,0)=f(x)\end{cases}\qquad \begin{cases}u_t=ku_{xx}\\ u_x(0,t)=0\\ u(x,0)=f(x)\end{cases}$$

extend $f_O(x)=\begin{cases}f(x)\\ 0\\ -f(-x)\end{cases}$: extend $f_E(x)=\begin{cases}f(x)\\ f(-x)\end{cases}$

$$\Rightarrow\begin{cases}u_t=ku_{xx}\\ u(x,0)=f_O(x)\end{cases}\qquad \Rightarrow\begin{cases}u_t=ku_{xx}\\ u(x,0)=f_E(x)\end{cases}$$

both $\Rightarrow$
$$u(x,t)=\frac{1}{\sqrt{4\pi kt}}\left(\int_{-\infty}^{0}+\int_0^{\infty}\right)e^{-\frac{(x-y)^2}{4kt}}f_{O/E}(y)dy$$
$$=\int_0^{\infty}\frac{\exp\left(-\frac{(x-y)^2}{4kt}\right)\pm\exp\left(-\frac{(x+y)^2}{4kt}\right)}{\sqrt{4\pi kt}}f(y)dy$$

1) Find the Fourier transform of $f(x)$, where $f(x)=1$ for $-2<x<2$ and $f(x)=0$ otherwise.
2) Find the Fourier transform of $f(x)$, where $f(x)=e^{-x^2/2}$.
3) Find the Fourier transform of $f(x)$, where $f(x)=e^{-(x-2)^2/2}$.
4) Find the Fourier transform of $f(x)$, where $f(x)=\frac{1}{1+(x-3)^2}$.

Sol (1)
$$\hat f(\mu)=\int_{-\infty}^{\infty}f(x)e^{-i\mu x}dx=\int_{-2}^{2}e^{-i\mu x}dx$$
$$=\left[\frac{e^{-i\mu x}}{-i\mu}\right]_{-2}^{2}=\frac{e^{-2i\mu}-e^{2i\mu}}{-i\mu}=\frac{-2i\sin(2\mu)}{-i\mu}=\frac{2\sin(2\mu)}{\mu}$$

(2) This is Gaussian function centered at 0, $\sigma=1$, multiplied by $\sqrt{2\pi}$
$$\Rightarrow \hat f(\mu)=\frac{\sqrt{2\pi}}{2\pi}e^{-\frac{\mu^2}{2}}=\frac{1}{\sqrt{2\pi}}e^{-\frac{\mu^2}{2}}$$

(3) This function is a Gaussian centered at 2, $\sigma=1$, multiplied by $\sqrt{2\pi}$
So $\hat f(\mu)=\frac{\sqrt{2\pi}}{2\pi}e^{-2i\mu}e^{-\mu^2/2}=\frac{1}{\sqrt{2\pi}}e^{-2i\mu}e^{-\frac{\mu^2}{2}}$

(4)
$$\hat f(\mu)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{1+(x-3)^2}e^{-i\mu x}dx=\frac{1}{2\pi}\cdot\pi e^{-|\mu|}e^{-3i\mu}$$
$$=\frac{1}{2}e^{-|\mu|}e^{-3i\mu}$$
by Lorentzian function centered at 3.

4.3 Fourier-Legendre series

$$\text{pks}\Rightarrow f(\theta)=\sum_{n=0}^{\infty}A_kP_k(\cos\theta),\ 0<\theta<\pi,\qquad f(s)=\sum_{n=0}^{\infty}A_kP_k(s),\ -1<s<1$$
$$A_k=\frac{2k+1}{2}\int_0^{\pi}f(\theta)P_k(\cos\theta)\sin\theta\,d\theta\qquad A_k=\frac{2k+1}{2}\int_0^{\pi}f(s)P_k(s)ds$$

ex compute F-L series of $f=\begin{cases}1, & s\in(0,1)\\ -1, & s\in(-1,0)\end{cases}$

$$f(s)=\sum_{k=0}^{\infty}a_kP_k(s),\quad a_k=\frac{2k+1}{2}\int_{-1}^{1}f(s)P_k(s)ds$$

Since f odd $\Rightarrow a_{2m}=0\Rightarrow f(s)=\sum_{m=0}^{\infty}a_{2m+1}P_{2m+1}(s)$

$$a_{2m+1}=\frac{4m+3}{2}\int_{-1}^{1}P_{2m+1}(s)ds=(4m+3)\int_0^1P_{2m+1}$$

$$\boxed{\int_0^1P_{2m+1}(s)ds}=\frac{1}{2^{2m+1}(2m+1)!}\int_0^1\left(\frac{d}{ds}\right)^{2m+1}(s^2-1)^{2m+1}ds$$
$$=\sim\left(\left(\frac{d}{ds}\right)^{2m}(s^2-1)^{2m+1}\right)\Big|_0^1=\sim\left(\frac{d}{ds}\right)^{2m}\Big|_{s=0}(s^2-1)^{2m+1}$$

note! $(a+b)^k=\sum_{l=0}^{k}\binom{k}{l}a^lb^{k-l}$, $\left(\frac{d}{ds}\right)^k\Big|_{s=0}s^l=l!\,\delta_{kl}$

$$\Rightarrow=\sim\left(\frac{d}{ds}\right)^{2m}\Big|_{s=0}\left(\sum_{l=0}^{2m+1}\binom{2m+1}{l}(-1)^{2m+1-l}s^{2l}\right)$$
$$=\sim\left(\frac{d}{ds}\right)^{2m}\Big|_{s=0}\binom{2m+1}{m}(-1)^{m+1}s^{2m}\quad \text{only this not 0.}$$

$\left(\frac{d}{ds}\right)^{2m}s^{2m}=(2m)!\Rightarrow$ it $=\frac{1}{2^{2m+1}(2m+1)!}\binom{2m+1}{m}(-1)^{m+1}(2m)!$

Set $f(s)=\sum_{i=0}^{2}a_iP_i(s)\Rightarrow a_0+a_1s+a_2\frac{3s^2-1}{2}=3s^2+4s+1$

$\Rightarrow\left(\frac{3a_2}{2}\right)s^2+a_1s+\left(a_0-\frac{a_2}{2}\right)=3s^2+4s+1$

$\Rightarrow a_0=2,\ a_1=4,\ a_2=2$

So $f(s)=2P_0(s)+4P_1(s)+2P_2(s)$

$=2+4r\cos\theta+r^2(3\cos^2\theta-1)$

eq on inf rod by Fourier trans

$$\begin{cases}u_t=ku_{xx}, & t>0,\ -\infty<x<\infty\\ u(x,0)=f(x), & -\infty<x<\infty\end{cases}\Rightarrow f(x)=\int_{-\infty}^{\infty}\tilde u(\mu,0)e^{i\mu x}d\mu$$

step1 Fourier trans:
$$u(x,t)=\int_{-\infty}^{\infty}\tilde u(\mu,t)e^{i\mu x}d\mu,\quad \tilde u(\mu,t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}u(x,t)e^{-i\mu x}dx$$
$$\Rightarrow u_t(x,t)=\int_{-\infty}^{\infty}\tilde u_t(\mu,t)e^{-i\mu x}d\mu,\quad u_{xx}(x,t)=\int_{-\infty}^{\infty}\tilde u(\mu,t)(-\mu^2)e^{i\mu x}$$
$\Rightarrow \hat u_{xx}=(i\mu)^2\hat u$ note: $\hat u_x=i\mu\hat u$
$$\Rightarrow\frac{\partial\hat u(\mu,t)}{\partial t}=-k\mu^2\hat u(\mu,t)$$
$$\Rightarrow\hat u(\mu,t)=e^{-k\mu^2t}\hat u(\mu,0)=e^{-k\mu^2t}\hat f(\mu)$$
$$\Rightarrow u(x,t)=\int_{-\infty}^{\infty}\hat f(\mu)e^{-k\mu^2t}e^{i\mu x}d\mu\qquad \frac{1}{2\pi}\int_{-\infty}^{\infty}f(y)e^{-i\mu y}dy$$
$$=\frac{1}{2\pi}\int_{-\infty}^{\infty}f(y)\int_{-\infty}^{\infty}\exp\left[-kt\left(\mu^2-\frac{2i(x-y)}{2kt}-\frac{(x-y)^2}{4k^2t^2}\right)-\frac{(x-y)^2}{4kt}\right]d\mu\qquad \left(\mu-\frac{i(x-y)}{2kt}\right)^2$$
$$=\sqrt{\frac{\pi}{kt}}\frac{1}{2\pi}\int_{-\infty}^{\infty}f(y)e^{-\frac{(x-y)^2}{4kt}}dy\qquad \text{note: }\int_{-\infty}^{\infty}e^{-a(x+b)^2}dx=\sqrt{\frac{\pi}{a}}$$
$$=\frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty}e^{-\frac{(x-y)^2}{4kt}}f(y)dy$$

5.4 d'Alembert's formula for wave eq.

$$\begin{cases}u_{tt}=c^2u_{xx}, & t>0,\ -\infty<x<\infty\\ u(x,0)=f_1(x), & -\infty<x<\infty\\ u_t(x,0)=f_2(x), & -\infty<x<\infty\end{cases}$$
$$\Rightarrow\begin{cases}\tilde u_{tt}+c^2\mu^2\tilde u=0\\ \tilde u(\mu,0)=\hat f_1(\mu)\\ \tilde u_t(\mu,0)=\hat f_2(\mu)\end{cases}$$

Since $\tilde u(\mu,t)=A(\mu)\cos(\mu ct)+B(\mu)\sin(\mu ct)$

By initial cond $A(\mu)=\hat f_1(\mu)$, $B(\mu)=\frac{\hat f_2(\mu)}{\mu c}$

$$u(x,t)=\int_{-\infty}^{\infty}\left[\hat f_1(\mu)\cos(\mu ct)+\hat f_2(\mu)\frac{\sin(\mu ct)}{\mu c}\right]e^{i\mu x}d\mu$$
$$=\frac{1}{2}\int_{-\infty}^{\infty}\hat f_1(\mu)\left[e^{i\mu(x+ct)}+e^{i\mu(x-ct)}\right]d\mu+\frac{1}{2c}\int_{-\infty}^{\infty}\hat f_2(\mu)\int_{x-ct}^{x+ct}e^{i\mu y}dy\,d\mu$$
$$=\frac{1}{2}[f_1(x+ct)+f_1(x-ct)]+\frac{1}{2c}\int_{x-ct}^{x+ct}f_2(y)dy$$

Question 5. Use d'Alembert's formula to solve the wave equation $u_{tt}=c^2u_{xx}$ with initial conditions $u(x,0)=0$ and $u_t(x,0)=4\cos 5x$.

$$u(x,t)=\frac{1}{2}[f_1(x+ct)+f_1(x-ct)]+\frac{1}{2c}\int_{x-ct}^{x+ct}f_2(y)dy$$
$$=\frac{1}{2c}\int_{x-ct}^{x+ct}4\cos 5y\,dy$$
$$=\frac{2}{5c}\int_{5x-5ct}^{5x+5ct}\cos z\,dz$$
$$=\frac{2}{5c}\left[\sin(5x+5ct)-\sin(5x-5ct)\right]$$
$$=\frac{2}{5c}\left(2\cos\left(\frac{10x}{2}\right)\sin\left(\frac{10ct}{2}\right)\right)=\frac{4}{5c}\cos 5x\sin 5ct$$

4.4 Laplace eq in 3D ($0\le r<a$, $0\le\alpha\le\pi$)

$$\begin{cases}\Delta u=0\Rightarrow\frac{1}{r^2}(r^2u_r)_r+\frac{1}{r^2\sin\theta}(\sin\theta u_\theta)_\theta=0\\ u(a,\alpha)=G(\alpha):\begin{cases}0, & 0<\alpha<\frac{\pi}{2}\\ 1, & \frac{\pi}{2}<\alpha<\pi\end{cases}\end{cases}$$

$u=R\Theta$, $\frac{(r^2R')'}{R}=-\frac{(\sin\alpha\,\Theta')'}{\sin\alpha\,\Theta}=\mu$

$$\Rightarrow\begin{cases}(\sin\alpha\,\Theta')'+\mu\sin\alpha\,\Theta=0, & \Theta(0),\Theta(\pi)<\infty\quad ①\\ (r^2R')'-\mu R=0 & ②\end{cases}$$

change of var: let $r=e^s\Rightarrow$ ② becomes $R''(s)+R'(s)-\mu R(s)=0$

The characteristic eq: $x^2+x-\mu=0\Rightarrow x=\frac{-1\pm\sqrt{1+4\mu}}{2}$

So sol: $R=Ae^{x_+s}+Be^{x_-s}=Ar^{x_+}+Br^{x_-}$

$x_\pm$ are required to be integers to make it smooth

$\Rightarrow\mu=k(k+1)$, $x_+=k$, $x_-=-k-1$ Finiteness gives $B=0$ otherwise $r^{-(k+1)}\to\infty$ as $r\to 0$

And $\Theta(\alpha)=P_k(\cos\alpha)$

Therefore $u(r,\alpha)=\sum_{k=0}^{\infty}A_kr^kP_k(\cos\alpha)$

$u(a,\alpha)=G(\alpha)\Rightarrow G(\alpha)=\sum_{k=0}^{\infty}A_ka^kP_k(\cos\alpha)$

$$\Rightarrow A_ka^k=\frac{2k+1}{2a^k}\int_0^{\pi}G(\alpha)P_k(\cos\alpha)\sin\theta\,d\theta$$
$$=\frac{2k+1}{2a^k}\int_{-1}^{1}G(s)P_k(s)ds=\frac{2k+1}{2a^k}\int_0^1P_k(s)ds$$

by Rodrigue,
$$\int_0^1P_k(s)ds=\frac{1}{2^kk!}\int_0^1\left(\frac{d}{ds}\right)^k(s^2-1)^kds$$
$$=\frac{1}{2^kk!}\left(\frac{d}{ds}\right)^{k-1}\Big|_{s=0}(s^2-1)^k$$
$$=\frac{1}{2^kk!}\left(\frac{d}{ds}\right)^{k-1}\Big|_{s=0}\sum_{l=0}^{k}\binom{k}{l}(-1)^{k-l}s^{2l}$$

Since $\left(\frac{d}{ds}\right)^k\Big|_{s=0}s^l=l!\,\delta_{kl}$

$\Rightarrow\left(\frac{d}{ds}\right)^{k-1}\Big|_{s=0}s^{2l}=l!\,\delta_{k-1,2l}\Rightarrow$ if $k=2m$, all vanish; if $k=2m+1$, only m-term supports

$$\Rightarrow\int_0^1P_{2m+1}(s)ds=\frac{1}{2^{2m+1}(2m+1)}\binom{2m+1}{m}(-1)^{m+1}$$
$$\Rightarrow u(r,\theta)=\sum_{m=0}^{\infty}\frac{(-1)^{m+1}(4m+3)}{2^{2m+1}(2m+1)}\binom{2m+1}{m}\left(\frac{r}{a}\right)^{2m+1}P_{2m+1}(s)$$

变体

$$\begin{cases}u_t=ku_{xx}+Du_x, & t>0,\ -\infty<x<\infty\\ u(x,0)=f(x), & -\infty<x<\infty\end{cases}$$

$$\hat u_t(\mu,t)=k(i\mu)^2\hat u(\mu,t)+Di\mu\hat u(\mu,t)$$
$$=(-k\mu^2+Di\mu)\hat u(\mu,t)$$
$$\hat u(\mu,t)=e^{(-k\mu^2+Di\mu)t}\hat u(\mu,0)=e^{(-k\mu^2+Di\mu)t}\hat f(\mu)$$
$$\Rightarrow u(x,t)=\int_{-\infty}^{\infty}e^{(-k\mu^2+Di\mu)t+i\mu x}\hat f(\mu)d\mu$$
$$=\frac{1}{2\pi}\int_{\mathbb{R}}e^{-k\mu^2t+i(Dt+x-y)\mu}d\mu\,f(y)dy$$
$$-k\mu^2t+i(Dt+x-y)\mu=-kt\left(\mu^2-\frac{i}{kt}(Dt+x-y)\mu\right)$$
$$=-kt\left(\mu-\frac{i}{2kt}(Dt+x-y)\right)^2-\frac{1}{4kt}(Dt+x-y)^2$$
$$\Rightarrow u(x,t)=\frac{1}{2\pi}\int\sqrt{\frac{\pi}{kt}}e^{-\frac{1}{4kt}(Dt+x-y)^2}f(y)dy,\qquad \frac{1}{2\pi}\sqrt{\frac{\pi}{kt}}=\frac{1}{\sqrt{4\pi kt}}$$

5.5 Green's function

$$G(x,y,t)=\frac{1}{\sqrt{4\pi kt}}e^{-\frac{(x-y)^2}{4kt}}$$

heat eq with $u(x,0)=f(x)$ soled with $u(x,t)=\int_{-\infty}^{\infty}G(x,y,t)f(y)dy$

Solving $\begin{cases}u_t-ku_{xx}=h(x,t), & 0<t<T,\ x\in\mathbb{R}\\ u(x,t)=0, & x\in\mathbb{R}\end{cases}$

$$u(x,t)=\int_{-\infty}^{\infty}G(x,y,t)f(y)dy+\int_0^t\int_{-\infty}^{\infty}G(x,y,t-s)h(y,s)dyds$$

Solving $\begin{cases}u_t-ku_{xx}=h(x,t), & 0<t<T,\ 0<x<\infty\\ ①\ u(0,t)=0\ (\text{or } u_x(0,t)=0\ ②)\\ u=f(x), & t=0,\ 0<x<\infty\end{cases}$

To solve for ① (Dirichlet's BC)

Define $G_D(x,y,t)=G(x,y,t)-G(x,-y,t)$

$$u(x,t)=\int_0^{\infty}G_D(x,y,t)f(y)dy+\int_0^t\int_0^{\infty}G_D(x,y,t-s)h(y,s)dyds$$

To solve for ② (Neumann's BC)

Define $G_N(x,y,t)=G(x,y,t)+G(x,-y,t)$

$\Rightarrow$ same except G_N 代替 G_D

Solving $\begin{cases}u_t-ku_{xx}=0, & 0<t<T,\ 0<x<a\\ u(0,t)=u(L,t)=0\\ u(x,0)=f(x)\end{cases}$

Define $G_L(x,y,t)=\sum_{m=-\infty}^{\infty}\left(G(x,y+2mL,t)-G(x,-y+(2m+2)L,t)\right)$

$$\Rightarrow u(x,t)=\int_0^LG_L(x,y,t)f(y)dy$$

(if non-homo with $h(x,t)$ then
$$u(x,t)=\int_0^LG_L(x,y,t)f(y)dy+\int_0^t\int_0^LG_L(x,y,t-s)h(y,s)dyds$$

$$\begin{cases}u_t=Ku_{xx}-au & t>0,\ -\infty<x<\infty,\\ u=f(x) & t=0,\ -\infty<x<\infty.\end{cases}$$

The Green's function for the equation without absorption:
$$G_0(x,x',t)=\frac{1}{\sqrt{4\pi kt}}\exp\left(-\frac{(x-x')^2}{4kt}\right)$$

The absorption term cause the sol. to decay exponentially over time
$$G_t=kG_{xx}-aG$$
$$\Rightarrow G(x,x',t)=e^{-at}G_0(x,x',t)$$
So $G(x,x',t)=e^{-at}\frac{1}{\sqrt{4\pi kt}}\exp\left(-\frac{(x-x')^2}{4kt}\right)$

So
$$u(x,t)=\int_{-\infty}^{\infty}G(x,x',t)f(x')dx'$$
$$=e^{-at}\frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty}\exp\left(-\frac{(x-x')^2}{4kt}\right)f(x')dx'$$

$$\begin{cases} x=\rho\cos\varphi \\ y=\rho\sin\varphi \\ z \end{cases} \Leftrightarrow \begin{cases}\rho=\sqrt{x^2+y^2} \\ \varphi=\arctan\frac{y}{x} \\ z\end{cases} \quad \partial_x=\cos\varphi\,\partial_\rho-\frac{\sin\varphi}{\rho}\partial_\varphi, \quad \partial_y=\sin\varphi\,\partial_\rho-\frac{\cos\varphi}{\rho}\partial_\varphi$$

$$\nabla^2 f=\Delta f=f_{xx}+f_{yy}=\frac{1}{\rho}(\rho f_\rho)_\rho+\frac{1}{\rho^2}f_{\varphi\varphi}$$

3.1 Solving homo Laplace

$$\begin{cases}\nabla^2 u=0, \ 0\le\rho\le R, \ \varphi\in[-\pi,\pi] \\ u(R,\varphi)=u_2(\varphi), \ \varphi\in[-\pi,\pi]\end{cases}$$

$u=R(\rho)\Phi(\varphi) \Rightarrow \frac{1}{\rho}(\rho R')'\Phi+\frac{1}{\rho^2}R\Phi''\Rightarrow \rho\frac{(\rho R')'}{R}=-\frac{\Phi''}{\Phi}=\lambda$

$$\begin{cases}\Phi''+\lambda\Phi=0, \ \Phi(-\pi)=\Phi(\pi), \ \Phi'(-\pi)=\Phi'(\pi) \\ \rho(\rho R')'-\lambda R=0\end{cases}$$

Let $\lambda=m^2 \Rightarrow \Phi_m(\varphi)=A_m\cos m\varphi+B_m\sin m\varphi$

let $\rho=e^s$

$\Rightarrow \rho(\rho R')'-m^2R=0 \Rightarrow R_m(s)=Ae^{ms}+Be^{-ms}=C_m\rho^m+D_m\rho^{-m}$

$$\Rightarrow u_m(\rho,\varphi)=\begin{cases}(C_m\rho^m+D_m\rho^{-m})(A_m\cos m\varphi+B_m\sin m\varphi), \ m\in\mathbb{N} \\ C_0+D_0\ln\rho, \ m=0 \ (\text{const})\end{cases}$$

$D_m=0$ since otherwise $|u|\to\infty$ as $\rho\to 0$, not smooth

$\Rightarrow u(\rho,\varphi)=\sum_{m=0}^\infty \rho^m(A_m\cos m\varphi+B_m\sin m\varphi)$

$u_2(\varphi)=u(R,\varphi)=A_0+\sum_{m=1}^\infty R^m(A_m\cos m\varphi+B_m\sin m\varphi)$

$\Rightarrow R^mA_m, R^mB_m$ are fourier coeff

solved by $R^mA_m=\frac{1}{\pi}\int_{-\pi}^{\pi}u_2\cos mx$, $B_m=\frac{1}{\pi}\int_{-\pi}^{\pi}u_2\sin mx$

non-homo case and Bessels

$$\begin{cases}\Delta u=-\lambda u \\ u(R,\varphi)=u_2(\varphi)\end{cases}$$

$-\lambda R\Phi=\frac{1}{\rho}(\rho R')'\Phi+\frac{1}{\rho^2}R\Phi''$

$-\lambda\rho^2=\frac{\rho(\rho R')'}{R}+\frac{\Phi''}{\Phi}\Rightarrow \frac{\rho(\rho R')'+\lambda\rho^2R}{R}=-\frac{\Phi''}{\Phi}=m^2$

$$\Rightarrow\begin{cases}\Phi''+m^2\Phi=0, \ \Phi(-\pi)=\Phi(\pi), \ \Phi'(-\pi)=\Phi'(\pi) \Rightarrow \Phi_m=A_m\cos m\varphi+B_m\sin m\varphi \\ \boxed{\rho(\rho R')'+(\lambda\rho^2-m^2)R=0}\end{cases}$$

Def Bessel's functions $J_m(x)$ are sols to

$x(xJ_m')'+(x^2-m^2)J_m=0$

(or say $J_m''+\frac{1}{x}J_m'+(1-\frac{m^2}{x^2})J_m=0$)

Let $r=\sqrt{\lambda}\rho \Rightarrow J_m(r)=R_m(\rho)$ is sol to it

$\Rightarrow u(\rho,\varphi)=\sum_{m=0}^\infty J_m(\sqrt{\lambda}\rho)(A_m\cos m\varphi+B_m\sin m\varphi)$

and $u_2(\varphi)=\sum_{m=0}^\infty J_m(\sqrt{\lambda}R)(A_m\cos m\varphi+B_m\sin m\varphi)$

$A_mJ_m(\sqrt{\lambda}R), B_mJ_m(\sqrt{\lambda}R)$ are fourier coeff of $u_2(\varphi)$

(solved by $\frac{1}{\pi}\int_{-\pi}^{\pi}\cdots$)

3.3 Integral formula

$$J_m(x)=\frac{1}{2\pi i^m}\int_{-\pi}^{\pi}e^{ix\cos\theta}e^{-im\theta}d\theta, \quad m\in\mathbb{N}\cup\{0\}$$

Pf $e^{i\rho\cos\varphi}$ is sol to $-\Delta u=u$

$\Rightarrow e^{i\rho\cos\varphi}=\sum_{m=0}^\infty J_m(\rho)(\alpha_m e^{im\varphi}+\beta_m e^{-im\varphi})$

$\Rightarrow \alpha_mJ_m(\rho)$ is Fourier coeff of $e^{i\rho\cos\varphi}$

$=\frac{1}{2\pi}\int_{-\pi}^{\pi}e^{i\rho\cos\varphi}e^{-im\varphi}d\varphi \Rightarrow$ proportional to J_m

Taylor expansion

$$J_m(x)=\sum_{n=0}^\infty\frac{(-1)^nx^{2n+m}}{2^{m+2n}(m+n)!n!}$$

3.5 Orthogonality

We know $R(x)=J_m(\sqrt{\lambda}x)$ is sol to $x(xR')'+(\lambda x^2-m^2)R=0$

But if together with B.C. $R(1)\cos\beta+R'(1)\cos\beta+R'(1)\sin\beta=0$

$\Rightarrow J_m(\sqrt{\lambda})\cos\beta+\sqrt{\lambda}J_m'(\sqrt{\lambda})\sin\beta=0$

Def $\{x_n^{(m)}\}_{n\in\mathbb{N}}$ are nonneg sol to $\boxed{J_m(x_n^{(m)})\cos\beta+x_n^{(m)}J_m'(x_n^{(m)})\sin\beta=0}$

$\Rightarrow$ ① $\int_0^1 J_m(xx_{n_1}^{(m)})J_m(xx_{n_2}^{(m)})x\,dx=0, \ \forall n_1\ne n_2$

② $\int_0^1 J_m(xx_n^{(m)})^2x\,dx=\begin{cases}\frac{1}{2}J_{m+1}(x_n^{(m)})^2, \ \beta=0 \\ \frac{x_n^2-m^2+\cot^2\beta}{2x_n^2}J_m(x_n^{(m)})^2, \ 0<\beta\le\frac{\pi}{2}\end{cases}$

3.4 Properties of J_m

(1) $J_m(0)=\begin{cases}1, \ m=0 \\ 0, \ m\ne 0\end{cases}$ (2) $J_m(x)=\frac{x}{2m}[J_{m-1}(x)+J_{m+1}(x)], \ m\in\mathbb{N}$

Pf (integral formula) **Pf** $J_{m-1}+J_{m+1}=\frac{1}{2\pi i^m}\left(i\int_{-\pi}^{\pi}e^{ix\cos\varphi}e^{-i(m-1)\varphi}-i\int_{-\pi}^{\pi}e^{ix\cos\varphi}e^{-i(m+1)\varphi}\right)$

(3) $J_m'(x)=\frac{1}{2}[J_{m-1}(x)-J_{m+1}(x)], \ m=0,1,\ldots$

Pf $\frac{1}{2\pi i^m}\int_{-\pi}^{\pi}\frac{d}{dx}(e^{ix\cos\varphi})e^{-im\varphi}d\varphi$

$=\frac{1}{2\pi i^m}\int_{-\pi}^{\pi}e^{ix\cos\varphi}\,i\frac{e^{i\varphi}+e^{-i\varphi}}{2}e^{-im\varphi}d\varphi$

$=\frac{1}{2}\left(\frac{1}{2\pi i^{m-1}}\int_{-\pi}^{\pi}e^{ix\cos\varphi}e^{-i(m-1)\varphi}d\varphi-\frac{1}{2\pi i^{m+1}}\int_{-\pi}^{\pi}e^{ix\cos\varphi}e^{-i(m+1)\varphi}d\varphi\right)$

$=\frac{1}{2}(J_{m-1}(x)-J_{m+1}(x))$ □

$=\frac{1}{2\pi i^m}\left(i\int_{-\pi}^{\pi}e^{ix\cos\varphi}(ie^{i\varphi}-ie^{-i\varphi})e^{-im\varphi}\right)$, $ie^{i\varphi}+ie^{-i\varphi}=2i\cos\varphi$...

$=\frac{2}{ix}\frac{1}{2\pi i^m}\int_{-\pi}^{\pi}(e^{ix\cos\varphi})'e^{-im\varphi}d\varphi$

$=\left(e^{ix\cos\varphi}e^{-im\varphi}\right)\Big|_{-\pi}^{\pi}-\int_{-\pi}^{\pi}e^{ix\cos\varphi}(e^{-im\varphi})'$, $=e^{-ix}-e^{-ix}=0$

$=\frac{2m}{x}\left(\frac{1}{2\pi i^m}\int_{-\pi}^{\pi}e^{ix\cos\varphi}e^{-im\varphi}\right)$ ▷

(4) $\frac{d}{dx}[x^mJ_m(x)]=x^mJ_{m-1}(x), \ m=1,2,\ldots$ (5) $\frac{d}{dx}[x^{-m}J_m(x)]=-x^{-m}J_{m+1}(x), \ m=0,1,\ldots$

Pf $mx^{m-1}J_m+x^mJ_m'$ $\qquad -mx^{-m-1}J_m+x^{-m}J_m'$

$=mx^{m-1}\frac{x}{2m}(J_{m-1}+J_{m+1})+\frac{x^m}{2}(J_{m-1}-J_{m+1})$ $\qquad =-mx^{-m-1}\frac{x}{2m}(J_{m-1}+J_{m+1})+\frac{x^{-m}}{2}(J_{m-1}-J_{m+1})$

$=x^mJ_{m-1}$

$$\begin{cases}x=r\sin\theta\cos\varphi \\ y=r\sin\theta\sin\varphi \\ z=r\cos\theta\end{cases}\Leftrightarrow\begin{cases}r=\sqrt{x^2+y^2+z^2} \\ \theta=\arctan\left(\frac{z}{\sqrt{x^2+y^2}}\right) \\ \varphi=\arctan\left(\frac{y}{x}\right)\end{cases}$$

$$\Delta f=f_{xx}+f_{yy}+f_{zz}=\boxed{\frac{1}{r^2}(r^2u_r)_r+\frac{1}{r^2\sin\theta}(\sin\theta\,u_\theta)_\theta+\frac{1}{r^2\sin^2\theta}u_{\varphi\varphi}}$$

Def Legendre polynomial

sol $\Theta_k(\theta)=P_k(\cos\theta)$ to $(\sin\alpha\,\Theta')'+(k(k+1)\sin\alpha)\Theta=0$ Legendre eq.

satisfying $\Theta(-\pi)=\Theta(\pi)$, $\Theta'(-\pi)=\Theta'(\pi)$ 称为 Legendre polynomial

Legendre eq is equiv to:

(2) $\Theta''+\cot\theta\,\Theta'+k(k+1)\Theta=0$

(3) $[(1-s^2)y']'+k(k+1)y=1$

(4) $(1-s^2)y''-2sy'+k(k+1)y=0$

(set $s=\cos\alpha$)

(1) ⟺ (2):

Pf $(\sin\theta\,\varphi'(\theta))'+k(k+1)\sin\theta\,\varphi(\theta)=0$

$\Leftrightarrow \cos\theta\,\varphi'(\theta)+\sin\theta\,\varphi''(\theta)+k(k+1)\sin\theta\,\varphi(\theta)=0$

$\Leftrightarrow \frac{\cos\theta}{\sin\theta}\varphi'(\theta)+\varphi''(\theta)+k(k+1)\varphi(\theta)=0$

$\Leftrightarrow \varphi''(\theta)+\cot\theta\,\varphi'(\theta)+k(k+1)\varphi(\theta)=0$

(3) ⟺ (4):

Pf $((1-s^2)y')'+k(k+1)y=0$

Since $((1-s^2)y')'=\frac{d}{ds}[(1-s^2)y']=-2sy'+(1-s^2)y''$

$\Leftrightarrow (1-s^2)y''-2sy'+k(k+1)y=0$

(2) ⟺ (3):

Pf Let $s=\cos\theta \Rightarrow \theta=\arccos s$, $y(s)=\varphi(\theta)$

$\frac{dy}{d\theta}=y'(s)(-\sin\theta)$, $\frac{d^2y}{d\theta^2}=-\cos\theta\,y'(s)-\sin\theta\frac{dy'(s)}{d\theta}$, $=y''(s)(-\sin\theta)$

$=-\cos\theta\,y'(s)+\sin^2\theta\,y''(s)$

So eq (2) ⟺

$y''(s)\sin^2\theta-y'(s)\cos\theta+\cot\theta(-\sin\theta\,y'(s))+k(k+1)y(s)=0$

$\Leftrightarrow y''(s)\sin^2(\theta)-2y'(s)\cos\theta+k(k+1)y(s)=0$

$\Leftrightarrow (1-s^2)y''(s)-2sy'(s)+k(k+1)y(s)=0$

note $((1-s^2)y')'=\frac{d}{ds}[(1-s^2)y]=-2sy'+(1-s^2)y''$

$\Leftrightarrow (1-s^2)y''-2sy'+k(k+1)y=0$

Prop Let $s=\cos\alpha \Rightarrow$

$P(s)$ is a $\boxed{\text{polynomial in } s \text{ of degree } k}$

Pf Suppose the Taylor series of $P(s)$ is $P(s)=\sum_{n=0}^\infty a_ns^n$

$\Rightarrow y'=\sum_{n=0}^\infty na_ns^{n-1}$

$\Rightarrow (1-s^2)y'=\sum_{n=0}^\infty na_ns^{n-1}-\sum_{n=0}^\infty na_ns^{n+1}=\sum_{n=-1}^\infty(n+2)a_{n+2}s^{n+1}+\sum_{n=0}^\infty(n+2)a_{n+2}-na_n]s^{n+1}$

$\Rightarrow [(1-s^2)y']'+k(k+1)y=\sum_{n=0}^\infty[(n+1)(n+2)a_{n+2}+(k(k+1)-n(n+1))a_n]s^n=0$

$\Rightarrow a_{n+2}=\frac{(n-k)(n+k+1)}{(n+1)(n+2)}a_n$ ✓

Fourier-Bessel expansion

of pw-sm on $x\in(0,1)$

Let J_m be a Bessel, and $\{x_n^{(m)}\}$ be nonneg sols of $J_m(x)\cos\beta+xJ_m'(x)\sin\beta=0$

$\Rightarrow f(x)=\sum_{n=1}^\infty A_nJ_m(xx_n^{(m)}), \ x\in(0,1)$ $\quad (\to\frac{1}{2}[f(x+0)+f(x-0)])$

$A_n=\frac{\int_0^1 f(x)J_m(xx_n^{(m)})x\,dx}{\int_0^1 J_m(xx_n^{(m)})^2x\,dx}$

ex $f(x)=1, \ 0<x<1$, $m=0, \beta=0$ (eg: $J_0(x)=0$)

$\Rightarrow 1=\sum_{n=1}^\infty A_nJ_0(xx_n^{(0)})$, $A_n=\frac{\int_0 J_0(xx_n)x\,dx}{\int_0 J_0(x_n)^2x\,dx}$

$(t=xx_n^{(0)})$ $\frac{\frac{1}{x_n^2}\int_0^{x_n}tJ_0(t)dt}{\int_0^1 J_0(xx_n^{(0)})^2x\,dx}=\frac{=(tJ_0(x))'}{=\frac{1}{2}J_1(x_n)^2}=\frac{2}{x_nJ_1(x_n)}$ by (4)

4.2 Rodrigue formula

$$P_k(s)=\frac{1}{2^kk!}\left(\frac{d}{ds}\right)^k(s^2-1)^k$$

Lemma if $j<k\Rightarrow \left(\frac{d}{ds}\right)^j(s^2-1)^k\Big|_{s=\pm1}=0$

Pf it is proportional to the jth Taylor coeff of $(s^2-1)^k$ at $s=1$

and $(s^2-1)^k=2^k(s-1)^k(1+\frac{s-1}{2})^k\Rightarrow$ jth Taylor coeff is 0

Lemma (important in computing Legendre)

$\left(\frac{d}{ds}\right)^k\Big|_{s=0}s^L=L!\,\delta_{kL}$

Prop Recursion formula for P_k

$nP_n(s)=(2n-1)sP_{n-1}(s)-(n-1)P_{n-2}(s)$,

$P_0(s)=1, \ P_1(s)=s$

Orthogonality

SL problem: $\begin{cases}(\sin\alpha\,\Theta')'+k(k+1)(\sin\alpha)\Theta=0 \\ \Theta(0), \Theta(\pi)<\infty\end{cases}$

Equivly $\begin{cases}((1-s^2)y')'+k(k+1)y=0 \\ y(-1), y(1)<\infty\end{cases}$ Sol: $y(s)=P_k(s)$

Applying SL Thm $\Rightarrow \int_{-1}^1 P_k(s)P_L(s)\,ds=\int_0^\pi P_k(\cos\theta)P_L(\cos\theta)\sin\theta\,d\theta$ $=\frac{2}{2k+1}\delta_{kL}$

Pf 0 part is by SL Thm

It suffices to show: $\int_{-1}^1(P_k(s))^2ds=\frac{2}{2k+1}$ ✓

$\int_{-1}^1(P_k(s))^2=\frac{1}{2^{2k}(k!)^2}\int_{-1}^1\left(\left(\frac{d}{ds}\right)^k(s^2-1)^k\right)^2ds$

$=\frac{(2k)!}{2^{2k}(k!)^2}\int_{-1}^1(s^2-1)^kds$ (by int by part k times)

note $\boxed{\left(\frac{d}{ds}\right)^{2k}(s^2-1)^k=(2k)!}$ ✓

and $\int_{-1}^1(s-1)^k(s+1)^k=\frac{(k!)^2}{(2k)!}\frac{2^{2k+1}}{2k+1}$ by int by part k times

$\Rightarrow \int_{-1}^1(P_k(s))^2ds=\frac{(2k)!}{2^{2k}(k!)^2}\frac{(k!)^2}{(2k)!}\frac{2^{2k+1}}{2k+1}=\frac{2}{2k+1}$

Note P_k is even function for even k

odd ~ for odd k ✓

vibrating membrane in cylinder

$$\begin{cases}u_{tt}=c^2\Delta u \\ u(\rho=a,\varphi,t)=0 \\ u(\rho,\varphi,0)=1 \\ u_t(\rho,\varphi,0)=0\end{cases}\quad u=R\Phi T \Rightarrow \frac{1}{c^2}\frac{T''}{T}=\frac{R''+\frac{1}{\rho}R'}{R}+\frac{1}{\rho^2}\frac{\Phi_{\varphi\varphi}}{\Phi} \Rightarrow \begin{cases}\frac{1}{c^2}\frac{T''}{T}=-\lambda \\ \frac{\rho^2R''+\rho R'}{R}-\lambda\rho^2=\mu \\ -\frac{\Phi_{\varphi\varphi}}{\Phi}=\mu\end{cases}$$

$=-\lambda$, $=\lambda$; $\frac{\rho^2R''+\rho R'}{R}-\lambda\rho^2=\frac{\Phi_{\varphi\varphi}}{\Phi}$, $=\mu$, $=-\mu$

$\Rightarrow$ ① $\Phi''+\mu\Phi=0, \ \Phi(-\pi)=\Phi(\pi)$

② $R''+\frac{1}{\rho}R'+(\lambda-\frac{\mu}{\rho^2})R=0$, $R(a)=0$ here

③ $T''+\lambda c^2T=0$

① $\Rightarrow \Phi_m=A\cos m\varphi+B\sin m\varphi$, $\mu_m=m^2$

(Fix m), put μ_m into ②: $R''+\frac{1}{\rho}R'+(\lambda-\frac{m^2}{\rho^2})R=0$

sol is exactly $R(\rho)=J_m(\rho\sqrt{\lambda})$

$R(a)=0\Rightarrow a\sqrt{\lambda}$ is picked from nonneg roots of $J_m(x)=0$ ($\beta=0$)

Let $\{x_n^{(m)}\}$ be nonneg roots of $J_m(x)=0$

$\Rightarrow \sqrt{\lambda_{mn}}=\frac{x_n^{(m)}}{a}$

③ $\Rightarrow T=C\cos c\sqrt{\lambda}t+D\sin c\sqrt{\lambda}t$

$\Rightarrow T_{mn}=C\cos\left(\frac{ctx_n^{(m)}}{a}\right)+D\sin\left(\frac{ctx_n^{(m)}}{a}\right)$

$\Rightarrow u_{mn}(\rho,\varphi,t)=J_m\left(\frac{\rho x_n^{(m)}}{a}\right)(A\cos m\varphi+B\sin m\varphi)\left(\cos\frac{ctx_n^{(m)}}{a}+D\sin\frac{ctx_n^{(m)}}{a}\right)$

$u_t(\rho,\varphi,0)=0\Rightarrow D=0$

$\Rightarrow u(\rho,\varphi,t)=\sum_{m=0}^\infty\sum_{n=1}^\infty\left[A_{mn}J_m\left(\frac{\rho x_n^{(m)}}{a}\right)\cos m\varphi+B_{mn}J_m\left(\frac{\rho x_n^{(m)}}{a}\right)\sin m\varphi\right]\cos\frac{ctx_n^{(m)}}{a}$

bring into $u(\rho,\varphi,0)=1\Rightarrow 1=\sum_{m=0}\sum_{n=1}(A_{mn}J_m(\cdots)\cos m\varphi+B_{mn}J_m(\cdots)\sin m\varphi)$

we multiply both sides by $J_m\left(\frac{\rho x_n^{(m)}}{a}\right)\cos m\varphi$, take integral over $(-\pi,\pi)$

Since $\boxed{\int_{-\pi}^{\pi}\cos mx\,dx=\int_{-\pi}^{\pi}\sin mx\,dx=0 \text{ if } m\ne 0}\Rightarrow A_{mn}, B_{mn}\ne 0$ iff $m=0$

$\Rightarrow u(\rho,\varphi,t)=\sum_{n=1}^\infty A_nJ_0\left(\frac{\rho x_n^{(0)}}{a}\right)\cos\frac{ctx_n^{(0)}}{a}$

Take into $u(\rho,\varphi,0)=1\Rightarrow 1=\sum_{n=1}^\infty A_nJ_0\left(\frac{\rho x_n^{(0)}}{a}\right)$

Since $1=2\sum_{n=1}^\infty\frac{J_0(x_n^{(0)})}{x_n^{(0)}J_1(x_n^{(0)})}\Rightarrow A_n=\frac{2}{x_n^{(0)}J_1(x_n^{(0)})}$

$\Rightarrow u=\sum_{n=1}^\infty\frac{2}{x_n^{(0)}J_1(x_n^{(0)})}J_0\left(\frac{\rho x_n}{a}\right)\cos\frac{ctx_n^{(0)}}{a}$

heat eq. in cylinder.

$$\begin{cases}u_t=k\nabla^2u+\sigma, \ t>0, 0\le\rho<a, -\pi\le\varphi\le\pi \\ u(a,\varphi,t)=T_1, \ t>0, -\pi\le\varphi\le\pi \\ u(\rho,\varphi,0)=T_0, \ 0\le\rho<a, -\pi\le\varphi\le\pi\end{cases}$$

Step $u=V(\rho,\varphi,t)+U(\rho)$

Find steady state sol $U(\rho)$ st $k\Delta U+\sigma=U_t=0$

$k\Delta U=k(U''+\frac{1}{\rho}U')=-\sigma \Rightarrow U''+\frac{1}{\rho}U'=-\frac{\sigma}{k}$

$\Rightarrow U'=-\frac{\sigma}{2k}\rho+\frac{\text{const}}{\rho}\Rightarrow U=-\frac{\sigma}{4k}\rho^2+C_1\ln\rho+C_2$

To bound $\ln\rho\Rightarrow C_1=0$

$U(a)=T_1\Rightarrow C_2=T_1+\frac{\sigma}{4k}a^2\Rightarrow U(\rho)=T_1+\frac{\sigma}{4k}(a^2-\rho^2)$

Then we solve for $V(\rho,\varphi,t)$: $\begin{cases}V_t=k\Delta V \\ V(a,\varphi,t)=0 \\ V(\rho,\varphi,0)=T_0-T_1\end{cases}$

By Separation: $\frac{1}{k}\frac{T'}{T}=-\lambda$, $\frac{\rho^2R''+\rho R'}{R}-\lambda\rho^2=\mu$, $\frac{\Phi''}{\Phi}=-\mu$

$$\Rightarrow\begin{cases}\Phi''+\mu\Phi=0, \ \Phi(-\pi)=\Phi(\pi), \ \Phi'(-\pi)=\Phi'(\pi) \\ R''+\frac{1}{\rho}R'+(\lambda-\frac{\mu}{\rho^2})R=0, \ R(a)=0 \\ T'+\lambda kT=0\Rightarrow T=\exp\left(-\frac{(x_n^{(0)})^2kt}{a^2}\right)\end{cases}$$

(Same as drumhead, except T' 替换 T'', T_0-T_1 替换 1, k 替换 c^2)

$\Rightarrow u=T_1+\frac{\sigma}{4k}(a^2-\rho^2)+\sum_{n=1}^\infty\frac{2(T_0-T_1)}{x_n^{(0)}J_1(x_n^{(0)})}J_0\left(\frac{\rho x_n^{(0)}}{a}\right)\exp\left(-\frac{(x_n^{(0)})^2kt}{a^2}\right)$

Question 1. Compute the following quantities.

1) Compute $P_1'(0), P_2'(0), P_3'(0), P_4'(0)$ by differentiating Legendre polynomials.
2) Compute $P_{2n+1}'(0)$ and $P_{2n}'(0) (n=0,1,2,\cdots)$ using Rodrigues' formula.
3) Compute $P_{2n}(0)$ using Rodrigues' formula.

Sol (1) $P_1(x)=x\Rightarrow P_1'(0)=1$

$P_2(x)=\frac{1}{2}(3x^2-1)\Rightarrow P_2'(0)=[3x]_{x=0}=0$

$P_3(x)=\frac{1}{2}(5x^3-3x)\Rightarrow P_3'(0)=\left[\frac{1}{2}(15x^2-3)\right]_{x=0}=-\frac{3}{2}$

$P_4(x)=\frac{1}{8}(35x^4-30x^2+3)\Rightarrow P_4'(0)=\left[\frac{1}{8}(140x^3-60x)\right]_{x=0}=0$

(2) P_{2n} is an even function $\Rightarrow P_{2n}'$ is odd $\Rightarrow P_{2n}'(0)=0$

$P_{2n+1}'(x)=\frac{1}{2^{2n+1}(2n+1)!}\frac{d}{dx}\left[\frac{d^{2n+1}}{dx^{2n+1}}(x^2-1)^{2n+1}\right]$

$=\frac{1}{2^{2n+1}(2n+1)!}\frac{d^{2n+2}}{dx^{2n+2}}\left(\sum_{k=0}^{2n+1}\binom{2n+1}{k}(-1)^{2n+1-k}x^{2k}\right)$

evaluated at $x=0\Rightarrow$ all term of deg $>2n+2$, and all terms of deg $<2n+2$ vanish

So $P_{2n+1}'(0)=\frac{1}{2^{2n+1}(2n+1)!}\frac{d^{2n+2}}{dx^{2n+2}}\binom{2n+1}{n+1}(-1)^nx^{2n+2}$

$=\frac{1}{2^{2n+1}(2n+1)!}\binom{2n+1}{n+1}(-1)^n(2n+2)!$

$=\frac{(-1)^n(2n+2)!}{2^{2n+1}(n+1)!n!}$

(3) (Same as (2)) $P_{2n}(0)=\frac{1}{2^{2n}(2n)!}\frac{d^{2n}}{dx^{2n}}\binom{2n}{n}(-1)^{2n-n}x^{2n}$

$=\frac{1}{2^{2n}(2n)!}\binom{2n}{n}(-1)^n(2n)!=\frac{(-1)^n(2n)!}{2^{2n}(n!)^2}$